

Medvale Internal Medicine

Chapter 3A - Gastroenterology and Hepatology Overview: Abdominal Pain, GI Bleeding, Diarrhea, Constipation, and Jaundice

Medvale Internal Medicine Project

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Chapter 3A. Gastroenterology and Hepatology Overview

Why this chapter matters clinically

Gastrointestinal and hepatobiliary complaints are among the most common reasons that adults enter emergency departments, inpatient services, and outpatient internal medicine clinics. The same symptom can represent a self-limited process, a surgical emergency, a malignancy, a medication effect, a vascular catastrophe, an autoimmune disease, or decompensated chronic organ failure. Abdominal pain can be viral gastroenteritis, perforated ulcer, pancreatitis, acute mesenteric ischemia, biliary obstruction, appendicitis, bowel obstruction, diabetic ketoacidosis, myocardial infarction, or ruptured abdominal aortic aneurysm. Diarrhea can be benign and short-lived, but it can also be inflammatory colitis, *Clostridioides difficile*, ischemic colitis, inflammatory bowel disease, malabsorption, medication toxicity, or endocrine disease. Jaundice may arise from hemolysis, hepatocellular injury, cholestasis, biliary obstruction, sepsis-associated cholestasis, inherited bilirubin disorders, or malignancy.

The internal medicine task is not to memorize every GI diagnosis as an isolated fact. The task is to rapidly sort unstable from stable patients, identify the anatomic compartment, determine whether the process is inflammatory, obstructive, ischemic, infectious, bleeding, malignant, metabolic, or medication-related, and then choose the next test or treatment. This overview chapter builds the framework for the organ-based chapters that follow. Later GI chapters will separately cover esophagus and stomach, small bowel and colon, pancreas and biliary tract, liver disease, inflammatory bowel disease, and GI malignancy. This chapter gives the problem-based map.

For graph RAG extraction, the important knowledge structure is: **symptom -> anatomic region -> mechanism -> dangerous diagnosis -> first test -> first treatment -> complication**. A learner who sees melena should connect the symptom to upper GI bleeding, hemodynamic resuscitation, restrictive transfusion thresholds, early endoscopy, peptic ulcer disease, varices, and medication risks. A learner who sees jaundice should connect bilirubin fractionation to hepatocellular injury, cholestatic injury, hemolysis, biliary obstruction, imaging with right upper quadrant ultrasound, and urgent ERCP when cholangitis is present.

Core clinical problem map

Presenting problem	First question	Dangerous diagnoses to exclude early	High-yield first tests	Initial management anchor
Abdominal pain	Is the patient unstable or peritoneal?	Perforation, bowel ischemia, ruptured AAA, ectopic pregnancy, obstruction, pancreatitis, cholangitis, appendicitis, MI	Vitals, CBC, CMP, lipase, lactate if ill, urinalysis, pregnancy test when relevant, ECG, CT abdomen/pelvis, RUQ ultrasound	IV access, fluids if hypovolemic, analgesia, antiemetics, antibiotics if sepsis/peritonitis, surgery consult when peritoneal or ischemic
GI bleeding	Upper, lower, or obscure?	Variceal bleed, brisk upper bleed, ischemic colitis, malignancy, diverticular hemorrhage, angiodysplasia	CBC, type and screen, coagulation tests, BUN/Cr, stool appearance, endoscopy/colonoscopy, CTA if brisk ongoing bleeding	Two large-bore IVs, resuscitation, transfusion if threshold met, PPI for suspected upper bleed, octreotide/antibiotics for suspected varices
Diarrhea	Acute vs chronic; inflammatory vs watery/fatty?	C. difficile, dysentery, severe dehydration, toxic megacolon, ischemic colitis, IBD flare, AIDS-related infection	Volume status, electrolytes, stool testing if severe/bloody/travel/immunocompromised/recurrent antibiotics, C. difficile test, inflammatory markers when chronic	Oral/IV rehydration, stop offending meds, avoid antimotility agents in dysentery/toxic megacolon, targeted antibiotics only when indicated
Constipation	Acute obstruction or chronic functional pattern?	Bowel obstruction, fecal impaction, colon cancer, volvulus, cauda equina, severe hypercalcemia	Medication review, rectal exam, electrolytes/TSH when indicated, abdominal imaging if obstruction	Fiber/PEG/stimulant options if uncomplicated, disimpaction if impacted, urgent imaging/surgery

Presenting problem	First question	Dangerous diagnoses to exclude early	High-yield first tests	Initial management anchor
			suspected, colon cancer evaluation when red flags	y if obstruction/peri tonitis
Jaundice	Unconjugated vs conjugated? Hepatocellular vs cholestatic?	Cholangitis, acute liver failure, pancreatic cancer, severe alcoholic hepatitis, viral hepatitis, hemolysis	Fractionated bilirubin, AST/ALT, alkaline phosphatase, INR, albumin, CBC, hemolysis labs, hepatitis serologies, RUQ ultrasound	Treat sepsis/obstruction urgently, stop hepatotoxins, supportive care, ERCP for obstructive cholangitis, transplant center if acute liver failure

Approach to abdominal pain

Step 1: decide whether the abdomen is dangerous before deciding what diagnosis is most likely

The first pass through abdominal pain is a stability screen. Hypotension, syncope, altered mental status, rigid abdomen, rebound tenderness, persistent tachycardia, fever with toxic appearance, gastrointestinal bleeding, severe dehydration, pregnancy, immunosuppression, advanced age, anticoagulation, or pain out of proportion to exam should move the patient into an emergency pathway. A soft abdomen and normal vital signs do not guarantee safety, but unstable vital signs or peritoneal signs change the tempo: resuscitate, image quickly, involve surgery or interventional services early, and avoid a slow outpatient-style workup.

Pain out of proportion to physical findings is a classic clue for mesenteric ischemia, especially in older adults with atrial fibrillation, atherosclerosis, low-flow states, recent vascular procedures, or hypercoagulable disease. Peritoneal signs suggest transmural injury, perforation, gangrenous bowel, advanced appendicitis, perforated diverticulitis, or complicated ischemia. Sudden maximal pain should raise concern for vascular catastrophe such as ruptured aneurysm, perforation, torsion, or ischemia. Pain that migrates, localizes, or becomes associated with guarding suggests progression from visceral to parietal peritoneal inflammation.

Step 2: use location as a starting map, not as proof

Abdominal pain localization is useful because visceral afferents map to embryologic and peritoneal territories, but early disease can be vague and older adults may have muted symptoms. Epigastric pain can be dyspepsia, peptic ulcer disease, pancreatitis, biliary colic,

inferior myocardial infarction, aortic pathology, or gastritis. Right upper quadrant pain suggests biliary colic, cholecystitis, cholangitis, hepatitis, hepatic abscess, congestive hepatopathy, or referred pneumonia. Right lower quadrant pain suggests appendicitis, ileitis, Crohn disease, cecal diverticulitis, ureteral stone, ovarian pathology, or hernia. Left lower quadrant pain in an older adult suggests diverticulitis but can also be ischemic colitis, colon cancer, constipation, or gynecologic disease. Diffuse pain suggests gastroenteritis, obstruction, ileus, peritonitis, diabetic ketoacidosis, mesenteric ischemia, inflammatory bowel disease, or medication/toxin effects.

Pain location	Common internal medicine diagnoses	Emergencies and mimics
Epigastric	GERD, gastritis, peptic ulcer disease, pancreatitis, biliary colic	MI, perforated ulcer, ruptured AAA, mesenteric ischemia
RUQ	Biliary colic, cholecystitis, hepatitis, congestive hepatopathy	Ascending cholangitis, hepatic abscess, Fitz-Hugh-Curtis syndrome, pneumonia
RLQ	Appendicitis, Crohn ileitis, cecal diverticulitis	Ovarian torsion, ectopic pregnancy, ureteral stone, strangulated hernia
LLQ	Diverticulitis, constipation, ischemic colitis	Colon cancer, volvulus, abscess, perforation
Diffuse	Gastroenteritis, ileus, obstruction, DKA, medication effect	Peritonitis, mesenteric ischemia, toxic megacolon, ruptured AAA

Step 3: identify the pain mechanism

Visceral pain is dull, crampy, poorly localized, and often accompanied by autonomic symptoms such as nausea, diaphoresis, or pallor. It arises from distention, stretch, inflammation, or ischemia of hollow organs. Parietal pain is sharper and more localized because the parietal peritoneum is somatically innervated. Referred pain occurs when visceral afferents converge with somatic dermatomes: diaphragmatic irritation can refer to the shoulder, biliary pain can radiate to the right scapula, and pancreatitis can radiate to the back. Colicky pain suggests intermittent obstruction of a hollow structure, such as biliary colic, ureteral stone, or bowel obstruction. Constant progressive pain suggests inflammation, ischemia, perforation, or malignancy.

A graph-RAG association should connect **colicky RUQ pain after fatty meals** to gallstones and cystic duct obstruction; **persistent RUQ pain with fever and Murphy sign** to acute cholecystitis; **RUQ pain with jaundice, fever, hypotension, or confusion** to cholangitis and urgent biliary drainage; **epigastric pain radiating to the back with elevated lipase** to pancreatitis; and **pain out of proportion plus elevated lactate** to mesenteric ischemia.

Step 4: choose the first test based on the suspected compartment

Right upper quadrant ultrasound is usually the first imaging test when gallbladder or biliary disease is suspected because it can identify gallstones, gallbladder wall thickening, pericholecystic fluid, sonographic Murphy sign, and biliary ductal dilation. CT abdomen/pelvis with IV contrast is often the workhorse for undifferentiated adult abdominal pain, diverticulitis, obstruction, perforation, appendicitis in many adults, ischemic complications, abscess, and malignancy. CT angiography is preferred when mesenteric ischemia or active arterial bleeding is suspected. Lipase helps support pancreatitis when the clinical syndrome fits, but mild lipase elevation can occur in renal failure, bowel ischemia, peptic ulcer disease, and other inflammatory states. ECG and troponin belong in the epigastric pain evaluation when cardiac ischemia is plausible.

Pregnancy testing is a safety test in reproductive-age patients because ectopic pregnancy changes the differential and imaging choices. Urinalysis can identify hematuria suggesting nephrolithiasis, pyuria suggesting urinary infection, or bilirubin/urobilinogen abnormalities that support hepatobiliary disease. Lactate is not a screening test that rules out ischemia when normal, but an elevated lactate in a toxic patient increases concern for shock, bowel ischemia, severe sepsis, or hypoperfusion.

Abdominal pain red flags

Red flags should be transformed into action rather than stored as trivia. Peritoneal signs trigger urgent surgical evaluation. Severe pain with atrial fibrillation or vascular disease triggers mesenteric ischemia evaluation. Fever plus jaundice triggers cholangitis evaluation. Back-radiating epigastric pain triggers pancreatitis and aortic differential thinking. Abdominal pain plus syncope or hypotension triggers hemorrhage, rupture, ectopic pregnancy, sepsis, and vascular catastrophe evaluation. Abdominal pain plus hematochezia in an older adult should raise ischemic colitis or brisk upper GI bleed in addition to diverticular bleeding.

Approach to GI bleeding

Stabilization comes before localization

The first question in GI bleeding is not whether the source is gastric, duodenal, colonic, or anorectal. The first question is whether the patient is perfusing. Tachycardia, hypotension, syncope, cool extremities, altered mental status, orthostasis, ongoing hematemesis, large-volume hematochezia, or severe anemia means the patient needs resuscitation while the diagnostic plan is being formed. Two large-bore peripheral IVs are preferred when possible; central access may be needed if peripheral access is poor, but central line placement should not delay resuscitation. Early type and screen or crossmatch, serial hemoglobin, platelet count, coagulation assessment, and medication review are core steps.

A restrictive transfusion strategy is commonly used for many hospitalized upper GI bleeding patients, with a hemoglobin threshold around 7 g/dL in stable patients, while active shock, massive hemorrhage, ongoing ischemia, or significant cardiovascular disease may require individualized decisions. The important educational point is that transfusion is not based on a

single hemoglobin value alone; it is based on hemoglobin, hemodynamics, rate of bleeding, perfusion, comorbid ischemic risk, and whether equilibration has occurred after fluid loss.

Upper versus lower bleeding patterns

Hematemesis strongly suggests an upper GI source proximal to the ligament of Treitz. Coffee-ground emesis suggests slower or partially digested upper GI bleeding. Melena usually indicates upper GI bleeding but can also occur with slow small bowel or right colon bleeding. Hematochezia usually suggests lower GI bleeding but can occur with brisk upper GI bleeding when transit is rapid. Maroon stool often suggests right colon or small bowel bleeding but is not perfectly specific. Occult blood loss presents as iron deficiency anemia, positive fecal occult blood testing, or unexplained microcytosis.

Finding	Most likely source pattern	Important caveat
Hematemesis	Upper GI bleed	Swallowed blood from epistaxis can mimic hematemesis
Coffee-ground emesis	Slower upper GI bleed	Does not prove bleeding has stopped
Melena	Upper GI bleed	Can occur from small bowel/right colon; iron and bismuth darken stool without bleeding
Hematochezia	Lower GI bleed	Brisk upper GI bleed can present with hematochezia and shock
Painless large-volume hematochezia	Diverticular bleeding, angiodysplasia	Malignancy and ischemia remain possible
Painful bloody diarrhea	Colitis: infectious, ischemic, inflammatory	Avoid reflexive ant motility therapy until dysentery/toxic megacolon risk is considered

Upper GI bleeding illness scripts

Peptic ulcer bleeding is associated with epigastric pain, NSAID use, aspirin, anticoagulants, *Helicobacter pylori*, critical illness, and prior ulcer disease. Mallory-Weiss tear follows retching or vomiting and causes mucosal laceration near the gastroesophageal junction. Esophagitis can bleed, especially in severe reflux, pill injury, infection, or immunocompromised states. Variceal bleeding occurs in portal hypertension and can present with massive hematemesis, shock, stigmata of chronic liver disease, thrombocytopenia, coagulopathy, ascites, or encephalopathy. Gastric cancer may cause chronic occult bleeding or overt bleeding with weight loss, early satiety, anorexia, or persistent vomiting.

When variceal bleeding is suspected, initial therapy includes resuscitation plus vasoactive therapy such as octreotide, antibiotic prophylaxis, and urgent endoscopic therapy. Non-variceal

suspected upper GI bleeding often receives proton pump inhibitor therapy and endoscopic evaluation. Endoscopy is both diagnostic and therapeutic: it can identify ulcers, visible vessels, active spurting, adherent clot, varices, erosions, malignancy, and vascular lesions, while allowing clipping, thermal therapy, injection therapy, band ligation, or other hemostatic interventions.

Lower GI bleeding illness scripts

Diverticular bleeding is classically painless hematochezia in an older adult and can be brisk. Angiodysplasia is associated with older age, chronic kidney disease, aortic stenosis, and recurrent painless bleeding. Ischemic colitis often causes crampy abdominal pain followed by bloody diarrhea, classically in watershed regions such as the splenic flexure or rectosigmoid junction. Infectious colitis produces diarrhea with fever, cramps, exposure history, travel, foodborne outbreak, or sick contacts. Inflammatory bowel disease causes chronic or recurrent bloody diarrhea, urgency, abdominal pain, weight loss, anemia, and extraintestinal manifestations. Colon cancer may cause occult bleeding, iron deficiency anemia, change in stool caliber, weight loss, or overt bleeding.

Colonoscopy is the major diagnostic tool once the patient is stable and prepared. CT angiography is useful in active brisk bleeding because it can localize bleeding without bowel preparation and can direct angiography or embolization. Tagged red blood cell scans are sensitive for intermittent bleeding but less anatomically precise. Anoscopy or sigmoidoscopy may identify anorectal sources such as hemorrhoids or fissures, but these should not automatically explain anemia, melena, or significant bleeding without an appropriate evaluation.

Risk stratification logic

Risk scores are not substitutes for clinical judgment, but they structure disposition decisions. In upper GI bleeding, a very low Glasgow-Blatchford score can identify some patients who may be safely discharged with close outpatient follow-up. Most patients with hemodynamic instability, active bleeding, significant anemia, liver disease, anticoagulation, comorbid disease, or concerning symptoms require hospital-level care. For graph extraction, connect **risk score** to **site-of-care decision**, not to a diagnosis.

Approach to diarrhea

First split: acute, persistent, or chronic

Acute diarrhea lasts less than 14 days, persistent diarrhea lasts 14-30 days, and chronic diarrhea lasts more than 4 weeks. Most acute watery diarrhea is infectious or toxin-mediated and self-limited, but severe dehydration, bloody stool, fever, severe abdominal pain, immunocompromise, older age, recent antibiotics, recent hospitalization, outbreak exposure, or travel changes the workup. Chronic diarrhea requires a different framework because the probability shifts toward inflammatory bowel disease, malabsorption, medication effects, endocrine disorders, bile acid diarrhea, microscopic colitis, chronic infection, pancreatic insufficiency, celiac disease, irritable bowel syndrome, and malignancy.

Second split: watery, inflammatory, or fatty

Watery diarrhea can be osmotic, secretory, functional, medication-related, endocrine, bile acid-mediated, or infectious. Osmotic diarrhea improves with fasting because unabsorbed solutes draw water into the lumen; examples include lactose intolerance, sorbitol, magnesium-containing antacids, and some malabsorptive states. Secretory diarrhea continues during fasting and can cause large-volume stool; examples include cholera-like toxin syndromes, bile acid diarrhea, microscopic colitis, some neuroendocrine tumors, and medication effects. Functional diarrhea and irritable bowel syndrome usually lack nocturnal symptoms, weight loss, blood, fever, severe anemia, or objective inflammation.

Inflammatory diarrhea is suggested by blood, pus, fever, tenesmus, severe pain, elevated fecal leukocytes/lactoferrin/calprotectin, or systemic toxicity. Causes include invasive bacterial infections, *C. difficile*, ischemic colitis, inflammatory bowel disease, radiation colitis, and malignancy. Fatty diarrhea or steatorrhea suggests malabsorption or maldigestion and is associated with bulky, greasy, foul-smelling, floating stools, weight loss, fat-soluble vitamin deficiency, and nutritional deficiencies. Causes include celiac disease, chronic pancreatitis, pancreatic exocrine insufficiency, small intestinal bacterial overgrowth, giardiasis, Crohn disease involving small bowel, and prior intestinal surgery.

Diarrhea type	Clues	Mechanism	Representative diagnoses
Watery osmotic	Improves with fasting; bloating; relation to foods or supplements	Nonabsorbed solutes pull water into lumen	Lactose intolerance, sorbitol, magnesium, carbohydrate malabsorption
Watery secretory	Persists with fasting; nocturnal; high volume	Active secretion or impaired absorption	Bile acid diarrhea, microscopic colitis, cholera-like toxins, endocrine tumors, medications
Inflammatory	Blood, fever, tenesmus, pain, elevated fecal calprotectin	Mucosal injury and inflammation	IBD, invasive infection, <i>C. difficile</i> , ischemic colitis
Fatty	Greasy bulky stool, weight loss, vitamin deficiency	Maldigestion or malabsorption of fat	Chronic pancreatitis, celiac disease, giardiasis, SIBO, Crohn disease

Acute diarrhea: when testing is useful

Mild acute watery diarrhea without red flags often needs no stool testing because it resolves with hydration and supportive care. Stool testing becomes more useful when diarrhea is severe, bloody, febrile, prolonged, associated with dehydration, associated with sepsis, occurs in immunocompromised patients, follows antibiotic exposure or hospitalization, or is linked to

travel/outbreak settings. *C. difficile* testing is appropriate when compatible diarrhea follows antibiotic use, healthcare exposure, inflammatory bowel disease flare, immunosuppression, or other risk factors. Testing formed stool for *C. difficile* is generally not helpful because colonization can be misclassified as infection.

Empiric antibiotics are not automatic. Many cases are viral or self-limited. Antibiotics can shorten selected bacterial diarrheal illnesses, but they may worsen outcomes in some toxin-producing infections and can increase adverse effects or resistance. Avoid antimotility agents such as loperamide when dysentery, high fever, toxic megacolon, or severe colitis is suspected. Rehydration is the central treatment. Oral rehydration is preferred when feasible; IV fluids are needed for severe dehydration, shock, inability to tolerate oral intake, or major electrolyte disturbances.

Chronic diarrhea: organized workup

Chronic diarrhea should begin with a medication and diet inventory. Common medication causes include metformin, magnesium, antibiotics, proton pump inhibitors, SSRIs, NSAIDs, colchicine, laxatives, acarbose, orlistat, and chemotherapy. Diet causes include lactose, fructose, sorbitol, caffeine, alcohol, and artificial sweeteners. Alarm features include weight loss, nocturnal diarrhea, blood, anemia, fever, family history of colon cancer or inflammatory bowel disease, onset after age 50, persistent severe pain, hypoalbuminemia, and abnormal inflammatory markers.

A common initial lab pattern includes CBC, electrolytes, renal function, albumin, thyroid testing when indicated, celiac serology when compatible, inflammatory markers or fecal calprotectin/lactoferrin when inflammatory disease is suspected, stool ova/parasites or *Giardia* testing when exposure fits, and colonoscopy with biopsies when alarm features, inflammatory markers, persistent unexplained diarrhea, or microscopic colitis risk are present. Microscopic colitis is important because the colon can look normal endoscopically; diagnosis requires biopsies.

Diarrhea graph associations

For graph RAG, link **bloody diarrhea + fever** to inflammatory diarrhea, stool testing, avoidance of loperamide, and differential diagnoses such as infectious colitis, IBD, ischemic colitis, and *C. difficile*. Link **chronic watery diarrhea in older woman on NSAID/PPI/SSRI** to microscopic colitis and colonoscopic biopsies. Link **bulky greasy stool + weight loss** to malabsorption and pancreatic/celiac/small bowel causes. Link **diarrhea after antibiotics** to *C. difficile*. Link **diarrhea with flushing/bronchospasm/right-sided valvular disease** to carcinoid syndrome as a less common but high-yield association.

Approach to constipation

Define the pattern and look for obstruction

Constipation can mean infrequent stools, hard stools, straining, incomplete evacuation, bloating, or need for manual maneuvers. A practical definition is fewer than 3 bowel

movements per week or difficult stool passage, but the patient's baseline matters. Acute constipation with severe abdominal pain, vomiting, obstipation, distension, high-pitched bowel sounds, hernia, prior abdominal surgery, cancer history, or peritoneal signs should be treated as possible obstruction until proven otherwise. Chronic constipation without red flags is usually approached through diet, medications, pelvic floor function, colonic transit, and metabolic contributors.

Common mechanisms

Normal-transit constipation is common and may overlap with irritable bowel syndrome. Slow-transit constipation reflects reduced colonic propulsion. Defecatory disorders result from pelvic floor dyssynergia or impaired rectal evacuation, often suggested by excessive straining, incomplete evacuation, digital maneuvers, or failure of standard laxatives. Secondary constipation can come from medications, metabolic disease, neurologic disease, structural lesions, pregnancy, immobility, dehydration, or endocrine disorders.

Medication causes are common and often missed. Opioids, anticholinergics, calcium channel blockers, iron, calcium supplements, aluminum antacids, tricyclic antidepressants, antihistamines, antipsychotics, diuretics causing dehydration, and some anti-Parkinson drugs can contribute. Metabolic causes include hypercalcemia, hypothyroidism, diabetes with autonomic neuropathy, hypokalemia, and chronic kidney disease. Neurologic causes include Parkinson disease, multiple sclerosis, spinal cord disease, stroke, and autonomic neuropathy.

Red flags that require diagnostic escalation

Constipation red flags include unintentional weight loss, iron deficiency anemia, rectal bleeding, positive fecal occult blood, family history of colorectal cancer, new onset in an older adult, change in stool caliber, persistent severe pain, fever, nocturnal symptoms, palpable mass, or failure to respond to appropriate therapy. Red flags do not diagnose cancer, but they change the workup toward colonoscopy or imaging rather than empiric laxatives alone. A rectal exam can identify fecal impaction, mass, fissure, hemorrhoids, blood, reduced tone, or dyssynergic pattern.

Constipation situation	Likely mechanism	Next step
Acute constipation + vomiting + distension	Obstruction or ileus	Abdominal imaging, surgical evaluation if obstruction/peritonitis
Chronic hard stools without alarm features	Functional/normal transit or diet-medication related	Fiber, hydration, polyethylene glycol, medication review
Constipation with manual maneuvers/incomplete evacuation	Defecatory disorder	Rectal exam, anorectal testing, pelvic floor therapy/biofeedback
Opioid-associated constipation	Reduced motility and secretion	Bowel regimen; consider peripherally acting mu-opioid antagonist when refractory

Constipation situation	Likely mechanism	Next step
New constipation + anemia/weight loss	Structural lesion or malignancy	Colonoscopy or appropriate colorectal evaluation

Treatment ladder

Initial treatment includes medication review, hydration when deficient, physical activity as tolerated, dietary fiber if appropriate, and osmotic laxatives such as polyethylene glycol. Fiber can worsen bloating in some patients and is not the right first step for suspected obstruction or severe impaction. Stimulant laxatives such as senna or bisacodyl can be used intermittently or as part of a regimen when osmotic therapy is insufficient. Secretagogues and prokinetic agents are options for chronic idiopathic constipation or constipation-predominant irritable bowel syndrome after basic measures fail. Fecal impaction requires disimpaction, enemas when appropriate, and prevention of recurrence.

Approach to jaundice and abnormal liver tests

Jaundice starts with bilirubin fractionation

Jaundice is visible yellowing of skin or sclera due to elevated bilirubin, often clinically apparent around a total bilirubin of 2-3 mg/dL, though this varies with lighting and skin tone. The first diagnostic split is unconjugated versus conjugated hyperbilirubinemia. Unconjugated bilirubin suggests overproduction, impaired hepatic uptake, or impaired conjugation. Causes include hemolysis, ineffective erythropoiesis, resorbing hematoma, Gilbert syndrome, Crigler-Najjar syndrome, and some medication effects. Conjugated bilirubin suggests hepatocellular dysfunction or impaired bile flow. Causes include viral hepatitis, alcoholic hepatitis, drug-induced liver injury, sepsis, cirrhosis, cholestatic liver disease, gallstones, strictures, pancreatic cancer, cholangiocarcinoma, and cholangitis.

A simple rule: **unconjugated bilirubin asks for hemolysis and conjugation physiology; conjugated bilirubin asks for liver injury and biliary obstruction.** Urine bilirubin supports conjugated hyperbilirubinemia because conjugated bilirubin is water-soluble. Dark urine and pale stools suggest impaired bile flow into the gut. Pruritus suggests cholestasis and bile salt accumulation. Fever with jaundice and right upper quadrant pain suggests ascending cholangitis and should not wait for a prolonged outpatient evaluation.

Hepatocellular, cholestatic, and mixed injury

Liver tests should be interpreted by pattern. Hepatocellular injury means AST and ALT are disproportionately elevated relative to alkaline phosphatase. Cholestatic injury means alkaline phosphatase is disproportionately elevated relative to AST and ALT. Mixed injury contains both. Bilirubin, INR, albumin, and platelets help estimate function and chronicity. AST and ALT are injury markers, not direct measures of liver synthetic function. INR and albumin better reflect synthetic function, although albumin changes slowly and can fall from inflammation, nephrotic syndrome, malnutrition, or protein-losing enteropathy.

Pattern	Lab clue	Common causes	Initial evaluation focus
Unconjugated hyperbilirubinemia	Indirect bilirubin predominates; no urine bilirubin	Hemolysis, Gilbert syndrome, ineffective erythropoiesis, impaired conjugation	CBC, reticulocytes, LDH, haptoglobin, smear, medication review
Hepatocellular injury	AST/ALT dominate	Viral hepatitis, alcohol, MASLD/MASH, ischemic hepatitis, autoimmune hepatitis, drug-induced liver injury	Medication/toxin review, viral serologies, alcohol history, metabolic risk, autoimmune tests when indicated
Cholestatic injury	ALP and direct bilirubin dominate	Gallstone obstruction, malignancy, PBC, PSC, infiltrative disease, drug injury	Confirm hepatic ALP source, RUQ ultrasound, biliary imaging, antimitochondrial antibody or MRCP when indicated
Mixed injury	AST/ALT and ALP both elevated	Drug-induced liver injury, viral hepatitis with cholestasis, biliary obstruction with hepatocyte injury	Medication review, imaging, targeted serologies

Important numeric anchors in liver disease

Extremely high AST/ALT values, often in the thousands, suggest ischemic hepatitis, acetaminophen toxicity, acute viral hepatitis, or severe toxin-mediated injury. Alcohol-associated hepatitis classically has AST greater than ALT, often with AST:ALT ratio greater than 2, but absolute AST is often below 300-500 U/L rather than in the several-thousand range. Alkaline phosphatase elevation should be confirmed as hepatic when uncertain using gamma-glutamyl transferase or other supporting data. INR elevation in acute hepatitis is a danger marker because it signals impaired synthetic function. Acute liver failure is defined by acute liver injury with coagulopathy and encephalopathy in a patient without preexisting cirrhosis; this requires urgent specialist and transplant-center-level thinking.

Jaundice illness scripts

Gilbert syndrome causes intermittent mild unconjugated hyperbilirubinemia, often triggered by fasting, illness, dehydration, or stress, with otherwise normal liver tests and no hemolysis. Hemolysis causes anemia, reticulocytosis, elevated LDH, low haptoglobin, indirect bilirubin elevation, and sometimes jaundice. Acute viral hepatitis causes malaise, anorexia, nausea, right upper quadrant discomfort, jaundice, dark urine, and hepatocellular enzyme elevation.

Cholelithiasis causes obstructive jaundice and ductal dilation. Ascending cholangitis causes fever, jaundice, right upper quadrant pain, and may progress to hypotension and confusion. Pancreatic head cancer causes progressive painless jaundice, weight loss, pale stools, dark urine, pruritus, and sometimes a palpable nontender gallbladder.

Imaging and next tests

Right upper quadrant ultrasound is the usual first imaging test for suspected biliary obstruction because it can identify gallstones, ductal dilation, gallbladder inflammation, masses, and ascites. If ducts are dilated, the problem is often extrahepatic obstruction and the next step may be MRCP, EUS, ERCP, or CT depending on stability and suspected cause. ERCP is therapeutic as well as diagnostic and is urgent when obstructive cholangitis is suspected. If ultrasound shows no obstruction and labs are hepatocellular, the evaluation shifts toward viral hepatitis, alcohol, metabolic-associated steatotic liver disease, drug-induced liver injury, autoimmune hepatitis, hemochromatosis, Wilson disease in younger patients, alpha-1 antitrypsin deficiency, and ischemic injury.

Nausea, vomiting, dysphagia, and weight loss as overview symptoms

Nausea and vomiting

Nausea and vomiting can arise from GI obstruction, gastroenteritis, gastroparesis, pancreatitis, biliary disease, pregnancy, medications, toxins, increased intracranial pressure, vestibular disorders, metabolic derangements, endocrine disease, myocardial infarction, uremia, diabetic ketoacidosis, adrenal insufficiency, and sepsis. Bile vomiting suggests obstruction distal to the ampulla of Vater or severe ileus. Feculent vomiting suggests distal obstruction or gastrocolic fistula. Projectile vomiting with neurologic symptoms suggests intracranial pathology. Vomiting with severe epigastric pain suggests pancreatitis, biliary disease, ulcer disease, or MI. Vomiting with abdominal distension and obstipation suggests bowel obstruction.

Evaluation depends on severity and context. Electrolytes can reveal hypokalemic, hypochloremic metabolic alkalosis from gastric losses. Pregnancy testing, glucose/ketones, renal function, medication review, lipase, liver tests, ECG, and imaging are selected based on the syndrome. Management includes volume and electrolyte replacement, antiemetics, bowel rest when obstruction is suspected, nasogastric decompression in selected obstruction/ileus cases, and treatment of the underlying cause.

Dysphagia

Dysphagia is divided into oropharyngeal and esophageal patterns. Oropharyngeal dysphagia causes difficulty initiating swallowing, coughing, choking, nasal regurgitation, aspiration, or wet voice. Causes include stroke, Parkinson disease, dementia, myasthenia gravis, structural lesions, and neuromuscular disease. Esophageal dysphagia causes food sticking after swallowing. Dysphagia to solids only suggests mechanical obstruction such as stricture, Schatzki ring, eosinophilic esophagitis, or malignancy. Dysphagia to solids and liquids from onset suggests motility disorder such as achalasia, diffuse esophageal spasm, or scleroderma esophagus.

Alarm features include weight loss, progressive symptoms, anemia, GI bleeding, odynophagia, age over 50 with new symptoms, smoking/alcohol risk, and persistent vomiting. Upper endoscopy is central for structural disease, biopsy, malignancy evaluation, eosinophilic esophagitis, and therapeutic dilation. Barium esophagram helps define rings, strictures, Zenker diverticulum, or motility patterns. Manometry is the key test for motility disorders after mechanical obstruction is excluded.

Unintentional weight loss and malnutrition

Weight loss in GI evaluation links many systems: malignancy, malabsorption, chronic infection, inflammatory bowel disease, chronic pancreatitis, celiac disease, endocrine disease, depression, medication effects, dysphagia, alcohol use disorder, cirrhosis, and systemic inflammatory disease. A graph RAG node for weight loss should connect to alarm features, cancer screening, chronic diarrhea, anemia, hypoalbuminemia, steatorrhea, dysphagia, early satiety, hepatosplenomegaly, lymphadenopathy, and constitutional symptoms.

Diagnostic tools in GI and hepatology

Labs

CBC identifies anemia, leukocytosis, thrombocytopenia, eosinophilia, and macrocytosis. Microcytic anemia suggests iron deficiency from chronic blood loss until proven otherwise in appropriate adults. Thrombocytopenia can suggest portal hypertension, hypersplenism, marrow disease, alcohol use, viral infection, or drug effects. CMP identifies electrolytes, renal function, glucose, albumin, calcium, and liver chemistries. Lipase is used for suspected pancreatitis; amylase is less specific. INR helps assess hepatic synthetic function and anticoagulation effect. Lactate is used when shock, ischemia, or severe sepsis is suspected.

Stool testing should be purposeful. Stool culture or multiplex PCR can identify bacterial pathogens when severe or inflammatory diarrhea is present. *C. difficile* testing should match the syndrome. Fecal calprotectin or lactoferrin suggests intestinal inflammation and can help distinguish inflammatory bowel disease from functional diarrhea in the right context. Fecal elastase supports pancreatic exocrine insufficiency. Fecal occult blood tests are screening tools and should not replace diagnostic evaluation of overt bleeding or iron deficiency anemia.

Endoscopy

Upper endoscopy evaluates dysphagia, upper GI bleeding, suspected ulcer complications, persistent vomiting, alarm symptoms in reflux/dyspepsia, celiac biopsy, esophagitis, varices, malignancy, and many therapeutic interventions. Colonoscopy evaluates lower GI bleeding, iron deficiency anemia, colorectal cancer screening, chronic diarrhea with alarm features, inflammatory bowel disease, microscopic colitis biopsies, and therapeutic polypectomy or hemostasis. Capsule endoscopy evaluates selected obscure small bowel bleeding after upper and lower sources are excluded. ERCP is mainly therapeutic: stone extraction, biliary drainage, stenting, sphincterotomy, and management of cholangitis or obstructive complications.

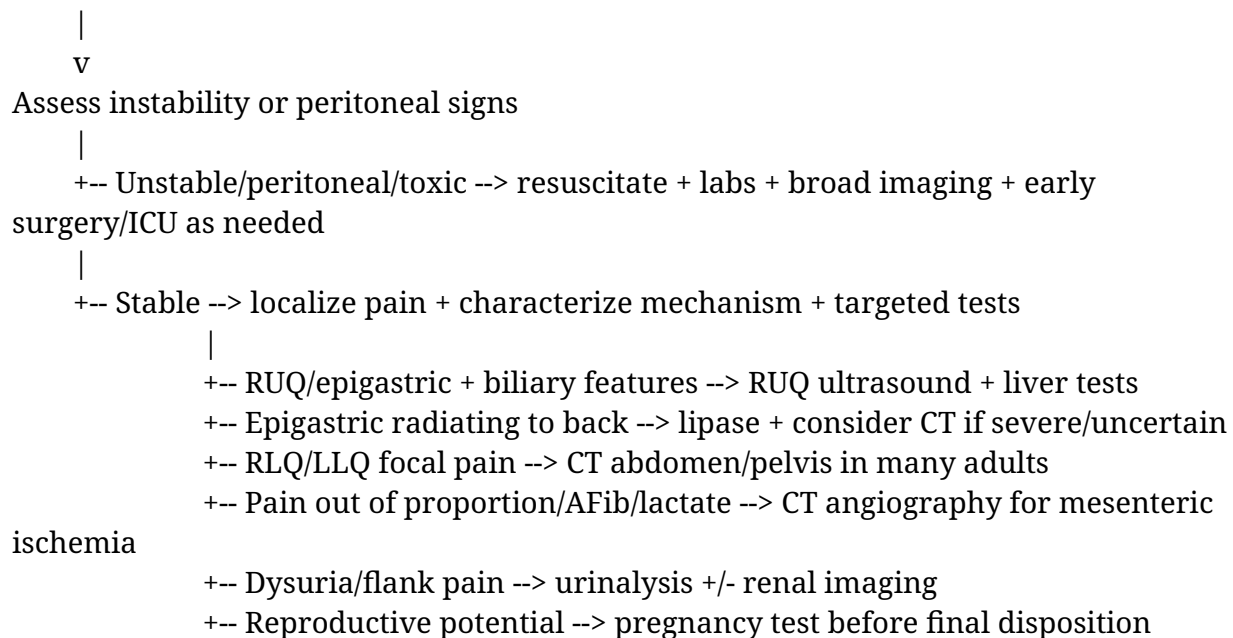
Imaging

CT abdomen/pelvis with IV contrast is useful for many acute abdominal processes: diverticulitis, appendicitis, obstruction, perforation, abscess, ischemic complications, pancreatitis severity, malignancy, and trauma. Ultrasound is first-line for gallbladder disease, biliary dilation, ascites, liver morphology, and vascular flow with Doppler when indicated. MRCP noninvasively evaluates the biliary tree and pancreatic duct. HIDA scan evaluates cystic duct obstruction when acute cholecystitis remains suspected despite nondiagnostic ultrasound. CT angiography identifies active bleeding and mesenteric ischemia. Plain radiographs have limited sensitivity but may help identify free air, obstruction patterns, stool burden in selected contexts, or toxic megacolon.

High-yield integrated algorithms

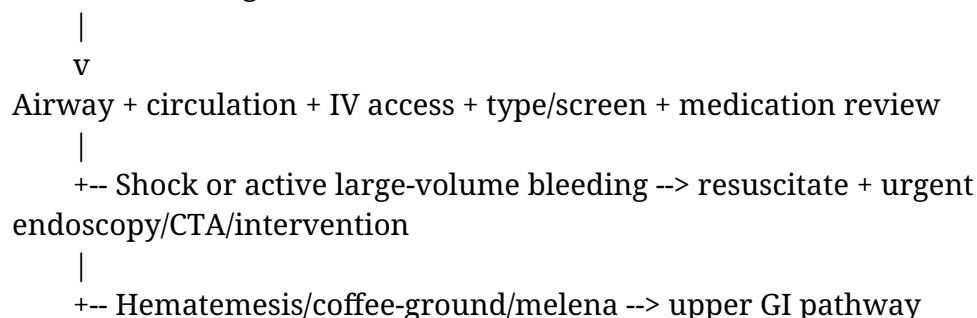
Algorithm 1: abdominal pain

Adult with abdominal pain



Algorithm 2: GI bleeding

Overt GI bleeding



- | PPI if non-variceal suspected
- | octreotide + antibiotics if variceal suspected
- | endoscopy after stabilization
- |
- +-- Hematochezia/maroon stool --> lower GI pathway
- consider brisk upper source if unstable
- colonoscopy when stable/prepped
- CTA if brisk ongoing bleeding

Algorithm 3: diarrhea

Diarrhea

- |
- v
- Acute (<14 days), persistent (14-30), or chronic (>4 weeks)
- |
- +-- Acute mild watery/no red flags --> oral rehydration + supportive care
- |
- +-- Bloody/severe/fever/travel/immunocompromised/recent antibiotics --> stool testing + targeted therapy
- |
- +-- Chronic --> classify watery vs inflammatory vs fatty
- medication/diet review
- CBC/CMP +/- TSH/celiac/fecal calprotectin/stool tests
- colonoscopy with biopsies if alarm/inflammation/unexplained

Algorithm 4: jaundice

Jaundice or elevated bilirubin

- |
- v
- Fractionate bilirubin
- |
- +-- Unconjugated predominant --> hemolysis labs + conjugation disorders + medication review
- |
- +-- Conjugated predominant --> AST/ALT/ALP pattern
- |
- +-- Hepatocellular --> viral, alcohol, metabolic, ischemic, autoimmune, drug/toxin workup
- +-- Cholestatic --> RUQ ultrasound for duct dilation/obstruction
- cholangitis signs? urgent antibiotics + ERCP drainage

High-yield numbers and thresholds

Numbers are useful when they trigger a clinical action. They are dangerous when memorized without context. The following are common internal medicine anchors that should be connected to decisions.

Number or threshold	Meaning	Clinical use
Hemoglobin around 7 g/dL	Common restrictive transfusion threshold in many stable hospitalized UGIB patients	Transfusion decision anchor; individualize in shock/ischemia/ongoing massive bleeding
Acute diarrhea <14 days	Acute category	Usually infectious/toxin/supportive unless red flags
Persistent diarrhea 14-30 days	Intermediate category	Consider persistent infection, postinfectious, medication, early chronic disease
Chronic diarrhea >4 weeks	Chronic category	Workup for inflammatory, malabsorptive, endocrine, medication, functional, malignancy causes
Constipation often <3 stools/week	Frequency criterion	Must also assess hard stools, straining, incomplete evacuation, baseline pattern
Visible jaundice often bilirubin 2-3 mg/dL	Clinical detection range	Prompts fractionated bilirubin and liver chemistry pattern analysis
AST:ALT >2 pattern	Alcohol-associated liver injury clue	Not diagnostic alone; absolute values and clinical context matter
ALT/AST in thousands	Severe acute hepatocellular injury pattern	Think ischemic hepatitis, acetaminophen/toxin, acute viral hepatitis
INR elevation in acute liver injury	Synthetic dysfunction	Danger marker; evaluate for acute liver failure if encephalopathy present
Charcot triad	Fever + RUQ pain + jaundice	Ascending cholangitis; urgent antibiotics and biliary drainage when obstructed
Reynolds pentad	Charcot triad + hypotension + confusion	Severe cholangitis/sepsis physiology
Lipase >=3x upper limit with compatible pain	Supports pancreatitis	Diagnosis usually requires compatible syndrome plus

Number or threshold	Meaning	Clinical use
Colon cancer screening age usually starts at 45 in average-risk adults in the US	Preventive care anchor	enzyme or imaging criteria Applies to average-risk; earlier for high-risk family/genetic/IBD histories

Common pitfalls

1. **Treating stable abdominal pain and unstable abdominal pain as the same problem.** The unstable patient needs resuscitation and rapid exclusion of surgical, vascular, septic, and hemorrhagic disease.
2. **Calling hematochezia lower GI bleeding without checking stability.** Brisk upper GI bleeding can present as hematochezia, especially with shock.
3. **Using stool occult blood testing to evaluate obvious bleeding.** Overt bleeding needs diagnostic evaluation, not a screening test.
4. **Giving loperamide to dysentery or severe colitis.** Antimotility agents can worsen toxic megacolon risk in inflammatory colitis syndromes.
5. **Assuming normal colonoscopy excludes microscopic colitis.** The mucosa may look normal; biopsies are required.
6. **Attributing iron deficiency anemia to hemorrhoids without appropriate evaluation.** Hemorrhoids can bleed, but they should not casually explain unexplained anemia or alarm features.
7. **Interpreting AST/ALT as liver function.** Injury markers are not the same as synthetic function. INR, albumin, bilirubin, encephalopathy, and platelets add functional and chronicity information.
8. **Missing medication causes.** NSAIDs, anticoagulants, metformin, PPIs, antibiotics, opioids, iron, anticholinergics, supplements, and acetaminophen are frequent GI/hepatic culprits.
9. **Forgetting non-GI mimics.** MI, pneumonia, DKA, adrenal crisis, testicular/ovarian pathology, urinary tract disease, and aortic disease can present with abdominal symptoms.
10. **Overusing antibiotics in diarrhea.** Many acute diarrheal illnesses are self-limited; antibiotics should be targeted to severity, host risk, pathogen risk, and public health context.

Practice questions

Question 1

A 74-year-old man with atrial fibrillation presents with sudden severe abdominal pain. He is tachycardic. His abdomen is soft with only mild diffuse tenderness. Lactate is mildly elevated. What diagnosis must be considered early, and what imaging best matches the concern?

Answer: Acute mesenteric ischemia should be considered because severe pain out of proportion to exam in a patient with atrial fibrillation suggests embolic or thrombotic intestinal ischemia. CT angiography is the key imaging study when suspicion is high. A normal early abdominal exam does not exclude ischemia.

Question 2

A 62-year-old woman presents with melena and lightheadedness. Blood pressure is 88/54 mmHg. What is the first management priority?

Answer: Stabilize circulation before definitive localization. Establish large-bore IV access, begin resuscitation, obtain type and screen/crossmatch and labs, review anticoagulants, and arrange urgent endoscopic evaluation after initial stabilization. Melena suggests an upper GI source, but shock changes the tempo of care.

Question 3

A 28-year-old traveler has 2 days of watery diarrhea without blood, fever, or severe pain and can drink fluids. What is the management principle?

Answer: Supportive care with oral rehydration is central. Stool testing and antibiotics are not needed for most mild acute watery diarrhea without red flags. The plan changes if fever, blood, severe dehydration, prolonged symptoms, immunocompromise, or outbreak concerns appear.

Question 4

A 68-year-old woman has 4 months of watery diarrhea, including nocturnal stools. Colonoscopy appears normal. She takes a proton pump inhibitor and an SSRI. What diagnosis remains possible, and why is biopsy important?

Answer: Microscopic colitis remains possible. The colon can appear normal on endoscopy, so random colonic biopsies are needed to diagnose collagenous or lymphocytic colitis.

Question 5

A patient has jaundice with direct bilirubin predominance, alkaline phosphatase elevation, fever, and right upper quadrant pain. What syndrome is this, and what is the management concept?

Answer: This is concerning for ascending cholangitis from infected biliary obstruction. Management requires resuscitation, antibiotics, and urgent biliary drainage, often by ERCP, rather than outpatient observation.

Question 6

A 53-year-old man with chronic alcohol use has AST 220 U/L, ALT 90 U/L, bilirubin elevation, and jaundice. How should the AST/ALT pattern be interpreted?

Answer: AST greater than ALT, especially a ratio greater than 2, supports alcohol-associated liver injury in the right clinical context. It is not diagnostic by itself. Synthetic function, infection, biliary obstruction, viral hepatitis, drug toxicity, and severity markers such as INR and encephalopathy still matter.

Question 7

A 71-year-old patient reports new constipation, iron deficiency anemia, and unintentional weight loss. What is the next-step logic?

Answer: This is not uncomplicated functional constipation. New constipation with anemia and weight loss is an alarm pattern requiring evaluation for structural disease, including colorectal cancer, usually with colonoscopy or appropriate imaging depending on the clinical situation.

Question 8

A patient with chronic greasy, floating stools has weight loss and low fat-soluble vitamin levels. What mechanism is most likely?

Answer: Fat malabsorption or maldigestion. Important causes include chronic pancreatitis with pancreatic exocrine insufficiency, celiac disease, giardiasis, small intestinal bacterial overgrowth, Crohn disease involving small bowel, and prior intestinal surgery.

Graph-RAG extraction map

The goal of this section is to make the chapter machine-readable at the concept level. Each row gives candidate nodes and relation types.

Source node	Relationship	Target node	Rationale
Abdominal pain	triaged by	Hemodynamic stability	Stable vs unstable determines urgency and site of care
Abdominal pain	localized to	RUQ, epigastric, RLQ, LLQ, diffuse	Location narrows anatomic differential
Pain out of proportion	suggests	Mesenteric ischemia	Visceral ischemia may precede peritoneal findings
RUQ pain + fever + jaundice	indicates	Ascending cholangitis	Charcot triad reflects infected biliary obstruction
Melena	suggests	Upper GI bleeding	Digested blood

Source node	Relationship	Target node	Rationale
			usually arises proximal to ligament of Treitz
Hematochezia + shock	may indicate	Brisk upper GI bleeding	Rapid transit can convert upper bleeding to red stool
Peptic ulcer disease	causes	Upper GI bleeding	Ulcer erosion can expose vessels
Portal hypertension	causes	Esophageal varices	Increased portal pressure creates portosystemic collaterals
Diverticulosis	causes	Painless hematochezia	Arterial bleeding can occur at diverticular vasa recta
Acute watery diarrhea	treated with	Rehydration	Volume replacement is central therapy
Bloody diarrhea	contraindicates careless use of	Antimotility agents	Severe inflammatory colitis may worsen with slowed transit
Chronic watery diarrhea	associated with	Microscopic colitis	Normal endoscopic appearance requires biopsies
Fatty diarrhea	suggests	Malabsorption or maldigestion	Greasy stool reflects impaired fat handling
Constipation	caused by	Opioids	Mu-opioid effects reduce gut motility and secretion
New constipation + anemia	suggests	Structural colon disease	Alarm features require diagnostic evaluation
Jaundice	evaluated by	Fractionated bilirubin	Direct vs indirect bilirubin separates mechanisms
Indirect bilirubin	associated with	Hemolysis	Heme breakdown increases unconjugated bilirubin load
Direct bilirubin + ALP elevation	suggests	Cholestasis or obstruction	Conjugated bilirubin rises when bile flow is impaired

Source node	Relationship	Target node	Rationale
INR elevation in acute hepatitis	indicates	Synthetic dysfunction	Coagulopathy is a severity marker
RUQ ultrasound	first-line for	Suspected biliary obstruction	Detects stones, duct dilation, and gallbladder inflammation
ERCP	treats	Obstructive cholangitis	Provides biliary drainage and stone/stent therapy

Selected clinical references used for accuracy checking

This Medvale chapter is original educational writing. The following sources were used to cross-check major clinical anchors and guideline-consistent phrasing:

1. American College of Gastroenterology. Clinical guidelines and patient-topic resources for gastrointestinal diseases.
2. Laine L, Barkun AN, Saltzman JR, Martel M, Leontiadis GI. ACG Clinical Guideline: Upper Gastrointestinal and Ulcer Bleeding. *American Journal of Gastroenterology*. 2021.
3. Kwo PY, Cohen SM, Lim JK. ACG Clinical Guideline: Evaluation of Abnormal Liver Chemistries. *American Journal of Gastroenterology*. 2017.
4. Riddle MS, DuPont HL, Connor BA. ACG Clinical Guideline: Diagnosis, Treatment, and Prevention of Acute Diarrheal Infections in Adults. *American Journal of Gastroenterology*. 2016.
5. American Gastroenterological Association and American College of Gastroenterology guidance on pharmacologic management of chronic idiopathic constipation in adults.